



TECHNICAL MEMORANDUM: LIGHTING PARAMETERS FOR BUILDING INFORMATION MODELING

AN AMERICAN NATIONAL STANDARD



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BUILDING INFORMATION MODELING
AN AMERICAN NATIONAL STANDARD**

Publication of this Technical Memorandum
has been approved by IES.

Suggestions for revisions
should be directed to IES.

**Prepared for IES
By the IES BIM Committee**



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1.0 Introduction and Scope

1.1 Introduction

The world is changing. To stay relevant, one must be willing to transform with it. The lighting design community is no different. The introduction of Building Information Modeling (BIM) is one such example. It is a data-centric, highly-integrated, and enduring-value approach to project delivery and management. For the entire lifecycle of a project – conceptualization through operations and even decommissioning – this methodology, with its associated processes and technologies, has the potential to assist all stakeholders involved in a project, saving time and money and improving the value of the final product.

One of the key features of BIM is the creation and exchange of digitally based information representing all aspects of the design, procurement, fabrication, and operations of a project and its components. The resulting set, or federated sets, of data is also known as a building information *model*. These models represent a “digital twin” of the larger project and its components, often in a three-dimensional format, with all the material, product, system, stakeholder, and process information attached. The models can vary in geometric and informational complexity and detail, dependent on the defined project goals and needs. From the creation and aggregation of all this geometry and information, numerous value-added workflows beyond formal design can be executed, such as automated product specification, code checking, building energy consumption simulation and analysis, lighting simulation and analysis, costing, procurement tracking, construction sequencing and simulation, and even automated operations monitoring and asset management).

Since 2008, the use of BIM technologies and processes has grown with nearly exponential speed throughout the design and construction fields. However, it has quickly become obvious that something is missing to achieve the full benefits of better, faster, cheaper and higher-value: standardization. This includes the standardization of processes, as well as the

standardization of information and its delivery for optimal (i.e., consistent, repeatable, reliable) use. One example of how the lack of standardized information affects BIM-based project delivery is the ad-hoc creation of content libraries – sometimes by designers and end-users, sometimes by product manufacturers, sometimes by BIM software vendors, sometimes by third parties.

The varied authors and resulting collections of geometry and information can contain incomplete, inconsistent, or even conflicting information, resulting in great detriment to optimal BIM methodology and high-quality project delivery. What was supposed to streamline project development is now a burden.

This Technical Memorandum recommends a standardization of parameters used in luminaire BIM content. The suggestions outlined in this document are intended to remedy synchronization issues between parameters, as well as ensure that lighting professionals are receiving the basic bits of information necessary to fulfilling their project’s needs, during the many different stages of development.

This document was developed by a group of professionals, known as the Illuminating Engineering Society BIM Committee, dedicated to the continued development and improvement of building information model management for the lighting industry.

1.2 Scope

This Technical Memorandum (TM) provides a recommended standardization of parameters attached to objects, object libraries, or parametric features that represent luminaires for use in many different types of BIM software. This ANSI/IES TM-32-24 revision provides specific parameter definitions and an associated shared parameters file that are recommended for use when developing lighting content for building information models. For each parameter, the following information is recommended:

- Parameter Grouping
- Parameter Name
- Description

- Tool Tip definition (for use specifically in Autodesk Revit*)
- Data Type
- GUID (for use specifically in Autodesk Revit*)

This TM does *not* include recommendations or explicit prescriptions for any of the values of the parameters, including geometric representation. The formatting of the associated data is the most important aspect to standardize to allow for compatibility across content generators and content users. Exact content, including geometry (2D and 3D) can vary according to project delivery requirements. Some geometric representations, such as 2D symbols for plans derived from the model, should follow other best-practice standards, such as ANSI/IES LP-7 Lighting Practice: Lighting Systems Contract Documents, the NECA 100-2013 NEIS Symbols for Electrical Construction Drawings (ANSI) or the NFPA 170 Standard for Fire Safety and Emergency Symbols. Requirements for 3D geometry may be based on the National BIM Standard – United States™ (NBIMS-US), Version 3, Chapter 2.7 – Level of Development Specification 2015,¹ or subsequent level of development (LOD) specification versions, which was produced and copyrighted by the American Institute of Architects as part of AIA Document G202™–2013.²

Parameters included and discussed in this TM are grouped into the following categories based on the aspect of the represented lighting element they define:

- General
- Photometrics
- Mounting
- Acoustics
- Mechanical
- Electrical
- Emergency
- Maintenance

* The IES does not endorse the use of any specific software. However, based on a survey of IES membership conducted in March 2021, it was determined that Revit is the BIM software predominantly used by IES members. Approximately 97% of IES members who use BIM software use Revit. Thus, the goal of the BIM Committee is to simplify adoption of this TM by including information specifically for the Revit platform.

For content creators, *all* parameters should be included, even with null values, so that the end-user may add the actual values during the design, procurement, installation, and operations process. Various information exchanges, where BIM data is transferred from one stakeholder and tool to another for a specific use—e.g., in lighting simulation and analysis—will determine which parameters are shared and when, according to BIM Project Execution Plans. This TM identifies all parameters that are possible but does not prescribe probability.

2.0 Definitions

For definitions related to the meaning of technical terms and parameters, the reader is referred to ANSI/IES LS-1-22.³ In some cases, parameters in this document may include in-place definitions and/or synonyms, to clarify meaning.

For a more comprehensive set of terms and definitions related to Building Information Modeling, the reader is referred to the National BIM Standard – United States™, Version 3, Chapter 3, Terms and Definitions, Sec. 3.4.1 Terms.¹

The following terms and definitions have been included for immediate reference and clarity regarding the methodological and technical contexts of BIM and the relationship to the content of this Technical Memorandum.

2.1 building information model (BIM)

The digital representation of physical and functional characteristics of a facility. As such, it serves as a shared knowledge resource for information about a facility, forming a reliable basis for decisions during its life cycle, from inception onward.

2.2 BIM deliverables

Information (a package of data or models in numerous formats) to be provided to another party in connection with a BIM-related service over the facility's lifecycle. When all parties agree on a common data format, or parameters, the information can be shared more seamlessly.

2.3 BIM implementation

1. The management of work processes and software tools to produce required project information.
2. In BIM software development: The incorporation of features in software, or the form those features take in the software.
3. In an IFC context: An application's capability to create, use, import, and export IFC project data.

2.4 BIM process map

A diagram showing how BIM will be applied on a project. The BIM project execution plan proposes two levels of process maps: a BIM overview map and detailed BIM use process maps that define associated activities and information exchanges.

2.5 BIM Project Execution Plan (BIM PxP Plan)

The plan that results from the BIM Project Execution Planning Procedure. The plan describes how BIM will be implemented and which goals or BIM Uses will be pursued.

2.6 BIM Project Execution Planning Procedure (PxP Procedure)

An approved NBIMS-US™ process for planning the execution of BIM on a project. It consists of four primary steps: 1) identify BIM Goals and BIM Uses, 2) design BIM Project Execution Process, 3) develop Information Exchanges, and 4) define supporting infrastructure for BIM Implementation. Guidelines and templates make these planning and design lifecycle processes reusable.

2.7 BIM use

A method of applying building information modeling during a facility's lifecycle to achieve one or more specific objectives. May also be referred to as a *BIM use case*.

2.8 digital twin

A digital twin is a digital representation of a real-world object or system. It is a virtual model that often mirrors the physical object or system in real time, collecting data and using it to simulate how the object or system would behave in different situations.

2.9 globally unique identifier (GUID)

A GUID is a 128-bit number used to identify information in computer systems. When generated according to the

standard methods, GUIDs are, for practical purposes, unique. Their uniqueness does not depend on a central registration authority or coordination between the parties generating them, unlike most other numbering schemes. While the probability that a GUID will be duplicated is not zero, it is generally considered close enough to zero to be negligible.

2.10 information accuracy

How close the information or information model is to an actual or planned result and associated requirements. Factors impacting accuracy include the level of detail and the level of precision expected at different phases of the BIM facility life cycle. The "build it first digitally" approach requires a very complete and very precise model for all systems included before the project enters physical construction. Information accuracy is developed during the design of a project.

2.11 information exchange

Packages of information passed from one party to another in a BIM process, or the act of passing such information. Can be a deliverable. Parties involved agree upon and understand what information content and format will be exchanged.

2.12 level of development (LOD)

The degree to which the element's geometry and attached information have been thought through; the degree to which project team members may rely on the information when using the model; the level of detail built into the BIM object.

2.13 parameter

In the context of a building information model (BIM) a parameter is a predefined container for information usually associated to geometry, and in some cases parametrically modifies the geometry. Restated, a parameter is a variable that is used to limit or control the behavior of inputted values. For example, a parameter defined to contain currency data would not allow letters to be entered.

2.14 shared parameter file

A "shared parameter file" is a resource used within the Autodesk Revit BIM-authoring platform. This file contains parameters associated with GUIDs which are

used in content (families), templates (project parameters), schedules, and tags. Using shared parameters for common data facilitates data sharing and integrity.

2.15 tool tip

Tooltips are a convenient way to get more information about something without having to look it up in a manual or search for it online. Depending on the BIM-authoring tool used, a tool tip can be associated with a parameter and visible to end-users while using a parameter during the design process.

3.0 Parameters

The following parameters are required to be included in any luminaire intended to be in compliance with this standard. The shared parameters file included in **Annex A** and found online⁴ can be downloaded to facilitate the incorporation of these parameters into BIM content.

3.1 General

Name	Description	Tool Tip	Data Type	GUID
Description	Description of product	Description or comments	Text	157f9768-92d6-4c2a-b6c6-4021de41bca4
Diameter	Overall Diameter of the housing of the luminaire or sensing device.	Overall Diameter of the housing of the luminaire or sensing device.	Length	c59a50f1-6e72-4981-ad58-2a65b8fbc846
Diameter of the Luminous Area	Diameter of the light circular shape aperture on the luminaire. Also indicated in the photometric file IES and LDT	Diameter of the light circular shape aperture on the luminaire. Also indicated in the photometric file IES and LDT	Length	59b50a71-0256-43f0-8982-c320c4ce2cda
Height	Height of the housing of the luminaire or sensing device. Corresponds to z-axis. This definition is regardless of orientation of luminaires.	Height of the housing of the luminaire or sensing device. Corresponds to z-axis. This definition is regardless of orientation of luminaires.	Length	8822b540-105f-47d1-847d-e3df372faa1e
Height of Luminous Area	The height of the luminous box of the luminaire. Also indicated in the photometric file.	The height of the luminous box of the luminaire. Also indicated in the photometric file.	Length	606a8c3b-8096-4d55-81e1-3ac6201886bf
Last Updated	Date BIM was last updated	Last file update time and date	Text	8350e45f-5dbb-4391-a637-620154f021c7
Length	Length of the housing of the luminaire or sensing device. Corresponds to y-axis. This definition is regardless of orientation of luminaires.	Length of the housing of the luminaire or sensing device. Corresponds to y-axis. This definition is regardless of orientation of luminaires.	Length	51782a3a-e0cb-45b4-86b1-2788dd5d6148
Length of Luminous Area	The length of the luminous box of the luminaire. Also indicated in the photometric file.	The length of the luminous box of the luminaire. Also indicated in the photometric file.	Length	1807c60b-35e5-4e55-897e-49570dee6b53

continued next page

3.1 General (continued)

Name	Description	Tool Tip	Data Type	GUID
Manufacturer	Manufacturer of product	Manufacturer of product	Text	9597d401-cfd4-4394-af6f-d84565b83b38
Manufacturer URL	URL of manufacturer website	Web URL of manufacturer website	URL	bddc75f4-8e27-4afc-add0-de7358bbf6d3
Product Name	Luminaire name or brand name of product	Luminaire name or brand name of product	Text	15f5ff4d-f917-45e4-ae77-5521e67f0a27
Width	Width of the housing of the luminaire or sensing device. Width is set parallel to the C0-C180 plane. Corresponds to the x-axis and with 0° horizontal of the Light Distribution Curve. This definition is regardless of orientation of luminaires.	Width of the housing of the luminaire or sensing device. Width is set parallel to the C0-C180 plane. Corresponds to the x-axis and with 0° horizontal of the Light Distribution Curve. This definition is regardless of orientation of luminaires.	Length	b3d433e9-f085-469f-810a-ead142e20f7e
Width of Luminous Area	The width is parallel to the C0-C180 plane. Aligned with the x-axis and with 0° horizontal.	Width is parallel to the C0-C180 plane. Aligned with the x-axis and with 0° horizontal.	Length	fecae275-f3d6-4f86-86df-7f957102c786

3.2 Photometric

Name	Description	Tool Tip	Data Type	GUID
Color Rendering Index (CRI)	The ability of a source to render color correctly.	The ability of a source to render color correctly.	Integer	acfa19a5-870c-4f39-8131-df68636c1cdc
Correlated Color Temperature	CCT. Correlated color temperature of the light source, in kelvins (e.g., 4500 K)	Measured in kelvins (K)	Color Temperature	Revit Built-in
Light loss factor	Multiplier used to predict the future performance (maintained luminous flux) relative to the initial properties of an installed lighting system.	Multiplier used to predict the future performance (maintained luminous flux) relative to the initial properties of an installed lighting system.	Number	Revit Built-in
Photometric Web File	Photometric test data for individual product.	Photometric data file.	.ies File	Revit Built-in

3.3 Mounting

Name	Description	Tool Tip	Data Type	GUID
Mounting type	Type of mounting used for the	The ability of a source to render color correctly.	Text	61767f07-f194-447c-b6fb-66f9285a7d39

3.4 Mechanical

Name	Description	Tool Tip	Data Type	GUID
Number of Light Outputs	Number of individual light sources (not number of diodes in LED, e.g., lamps in chandelier) - disassociate from GLDF	Number of plane surfaces with light output.	Integer	645a1d97-cd6f-4820-b1b4-ebe20ffe0e9c

3.5 Electrical

Name	Description	Tool Tip	Data Type	GUID
Apparent Load	Apparent power used by luminaire.	Also known as Input watts	Apparent Power	Revit Built-in
Dimmable	Determines whether the luminaire is dimmable.	Determines whether the luminaire is dimmable.	Yes/No	4d4881f3-5a77-41eb-a140-e03a297c2351
Dimming Method/ Protocol	Examples: Phase dimming, 0-10V, DMX.	Determines which dimmer type is applicable for the luminaire.	Text	b530286d-a1c4-4c8e-81e3-87b8075d96ad
Dimming range	Percentage	Determines which range of dimming is possible for luminaire.	Number	62f1c928-b7e3-496f-9e2d-26a6de767852
LED Module Replaceable	Yes/No - entire fixture does not need to be replaced.	LED module replaceable.	Yes/No	1ba13498-6fa6-44f8-989e-dff350b825ce
Light Source	Also known as "lamp".	Type of light source (e.g., fluorescent, LED, HID).	Text	3cd5049e-0b1e-4cd7-963c-6a59f1b87ee4
Rated Input Power	Input power consumed by the light source(s), control gear and control circuit in or associated with a luminaire including any standby power when the luminaire is turned on. Measured in volt-amps (VA).	Input power consumed by the light source(s), control gear and control circuit in or associated with a luminaire including any standby power when the luminaire is turned on.	Apparent power	c9846d7e-34ca-4f59-94ab-3452803f997e
Voltage Nominal	Nominal voltage at which the product operates normally as specified by the supplier or manufacturer.	Nominal voltage at which the product operates normally as specified by the supplier or manufacturer.	Electrical potential	7bcae613-49d5-40a3-a27e-8df1945ce73e

Additional Reading

International Standards Organization. ISO 19650-1:2018, *Organization and Digitization of Information About Buildings And Civil Engineering Works, Including Building Information Modelling (BIM) –Part 1: Concepts And Principles*. Geneva: ISO; 2018. (ICS 93 93.010).

Annex A Shared-Parameters File

A shared parameters file for use with third-party software is available for download at https://media.ies.org/docs/standards/TM-32-24_AnnexA.txt.⁴

This shared-parameters file includes both the parameters that are required for compliance with ANSI/IES TM-32-24 and the additional recommended parameters described in Annex B. In the shared-parameters file, the required parameters are organized into separate groups from the recommended parameters. The required parameter groups are designated by “ – Required” at the end of the group names. The recommended parameter groups are designated by “ – Recommended” at the end of the group names.

Annex B Additional Parameters

During the development of ANSI/IES TM-32-24, the IES BIM Committee developed an additional list of parameters that could be included in luminaire BIM content. While the inclusion of these parameters is not necessary for compliance with this standard, if these parameters are included, they are required to follow the format defined here and to use the listed GUID in order to allow for interoperability and data exchange.

In addition to providing parameter definitions, this annex maps the parameters defined in ANSI/IES TM-32-23 to other BIM standards and data formats used throughout the world, including the National Electrical Manufacturers Association (NEMA) standard BIM 100-2021,⁵ European Committee for Standardization Technical Specification CEN/TS 17623:2021,⁶ and the Global Lighting Data Format (GLDF).⁷

These additional parameters can be found online at https://media.ies.org/docs/standards/TM-32-24_AnnexB.xlsx.⁸

REFERENCES

- 1 National Institute of Building Sciences. National BIM Standard – United States, v3. Washington, DC; NIBS; 2015. Online: <https://www.nationalbimstandard.org/nbims-us-v3>. (Accessed 2024 Feb 6).
- 2 American Institute of Architects. G202-2013, Project Building Information Modeling Protocol Form. Washington, DC: AIA; 2013.
- 3 Illuminating Engineering Society. ANSI/IES LS-1-22, Lighting Science: Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2022.
- 4 ANSI/IES TM-32-24 Annex A. Online: https://media.ies.org/docs/standards/TM-32-24_AnnexA.txt. (Accessed 2024 May 7).
- 5 National Electrical Manufacturers Association. BIM 100-2021, BIM Data Requirements for Electrical Products in Support of Design, Construction, Operation, and Maintenance. Rosslyn, Virg.: NEMA; 2021.
- 6 European Committee for Standardization Technical Specification. PD CEN/TS 17623:2021, BIM Properties for lighting. Luminaires and sensing devices. Brussels, Belgium: CEN National Committees of the EU; 2021.
- 7 DIAL GmbH & RELUX Informatik AG. Global Lighting Data Format. Online: <https://glidf.io>. (Accessed 2023 Aug 30).
- 8 ANSI/IES TM-24-24 Annex B. Online: https://media.ies.org/docs/standards/TM-32-24_AnnexB.xlsx. (Accessed 2024 May 7).

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This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

- a) Proposed change accepted for public review without modification
- b) Proposed change accepted for public review with modification
- c) Proposed change accepted for further study
- d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

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- ☐ Change to read as follows
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Use underscore to show material to be added (added) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

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Select as applicable:

- ☐ Additional pages are attached. Number of additional pages: _____
- ☐ Attachments or referenced materials cited in this proposal accompany this proposed change.

Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:



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